



The Hotel School
Cornell
SC Johnson College of Business

Grading Analysis Report

By: Chris Jose

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Background

Over the past several semesters, I have worked closely with students while serving in a lead assistant role. Having recently been a student myself, I understand how grades impact motivation, confidence, and long term performance. I also understand how difficult it can be to identify issues early enough to meaningfully intervene.

As enrollment grew and the course evolved across multiple terms, I wanted a structured way to evaluate performance trends beyond what Canvas alone provides. My goal was simple: increase students' performance by understanding the data more clearly.

This project emerged from that need. It is both an operational improvement initiative and an analytical framework designed to support students more effectively over time

Introduction

This project builds a standardized, multi term grading warehouse and analytical framework using historical course data. The objective is to create consistency across semesters, validate grading structure, and identify patterns that can improve student outcomes.

Rather than treating each term independently, the system consolidates grade exports into a unified structure that allows:

- Cross term comparison
- Assignment level trend analysis
- Early identification of potential academic risk
- Operational insights for instructional planning

Grades are still delivered to students through Canvas. However, for analysis and quality assurance purposes, Canvas gradebook exports are extracted and processed through a structured ETL pipeline. The cleaned data is then stored in a database warehouse to enable repeatable, consistent analysis across semesters.

This approach ensures that grading data is not only recorded, but also understood.

Data Sources and Grade Delivery

Grades are delivered to students through Canvas, which serves as the official system of record. For analytical purposes, Canvas gradebook exports are downloaded in CSV format at the end of each term. These exports serve as the raw data source for this project.

Because assignment structures can vary slightly across semesters, term level datasets are harmonized into a consistent schema. This ensures that comparisons across terms reflect true performance differences rather than structural inconsistencies. The result is a standardized dataset where assignments, exams, and templates are comparable across multiple academic periods.

ETL and Database Warehouse

To support multi term analysis, a structured ETL process was implemented. Term level grade tables and assignment data are pulled from structured exports and local storage. Major assignment datasets are consolidated to support downstream scoring and comparative analysis.

Each semester's data is treated as an independent input source before being standardized. Raw point totals are normalized into percentage scores using max point mappings. Assignment names are standardized to remove term specific noise and inconsistencies. This ensures that features such as Prelim I, Final Exam, Excel Model, or Template assignments represent the same academic component across semesters. Assignments that were not offered consistently across terms are removed to preserve comparability. This prevents structural bias in cross term analysis.

The transformation stage focuses on accuracy, consistency, and comparability. After cleaning and standardization, the datasets are stored in a structured database warehouse.

A consolidated multi term table is built to support repeatable querying and consistent analytics. This allows term over term comparisons, segmentation analysis, and modeling to be performed on a stable, unified dataset rather than ad hoc spreadsheets. The warehouse serves as the analytical foundation for all subsequent evaluation

Analytics and Results

The analysis evaluates performance patterns across assignments, exams, templates, and time. The objective is not just to report averages, but to understand structure, stability, and variation across semesters. The figures below highlight cross term consistency, assignment level dispersion, and within term volatility

Average Score by Assignment

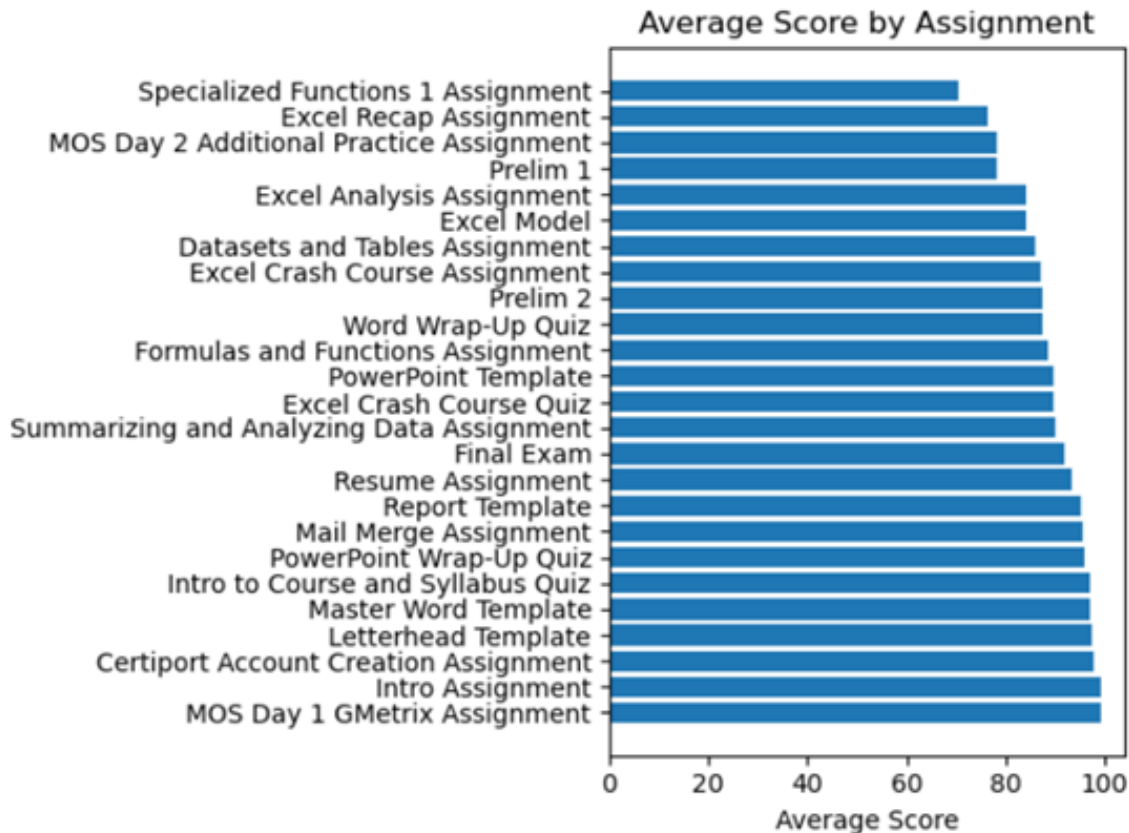


Figure 1: Average Score by Assignment

The assignment level distribution reveals a clear performance hierarchy.

Core introductory assignments and standardized templates cluster at the top of the range, with averages approaching 95 to 100. These include items such as Intro Assignment, MOS Day 1 GMetrix, Letterhead Template, and Master Word Template. These assignments demonstrate that students perform strongly when instructions are procedural and outcomes are well defined.

In contrast, more cognitively demanding or technical assignments show lower averages. Specialized Functions 1, Excel Recap, and MOS Day 2 Additional Practice sit noticeably lower. Prelim I also trends below Final Exam and several structured template assignments.

This pattern suggests three structural insights:

- Early conceptual Excel topics create more performance dispersion.
- Technical skill reinforcement assignments expose skill gaps more clearly than templates.
- Assessment difficulty increases when students must apply logic rather than follow structured formatting instructions.

Importantly, the overall distribution is still concentrated in the 80 to 100 range. This indicates strong aggregate performance, but the left tail of technical assignments is where academic risk becomes visible

Average Exam Scores by Term

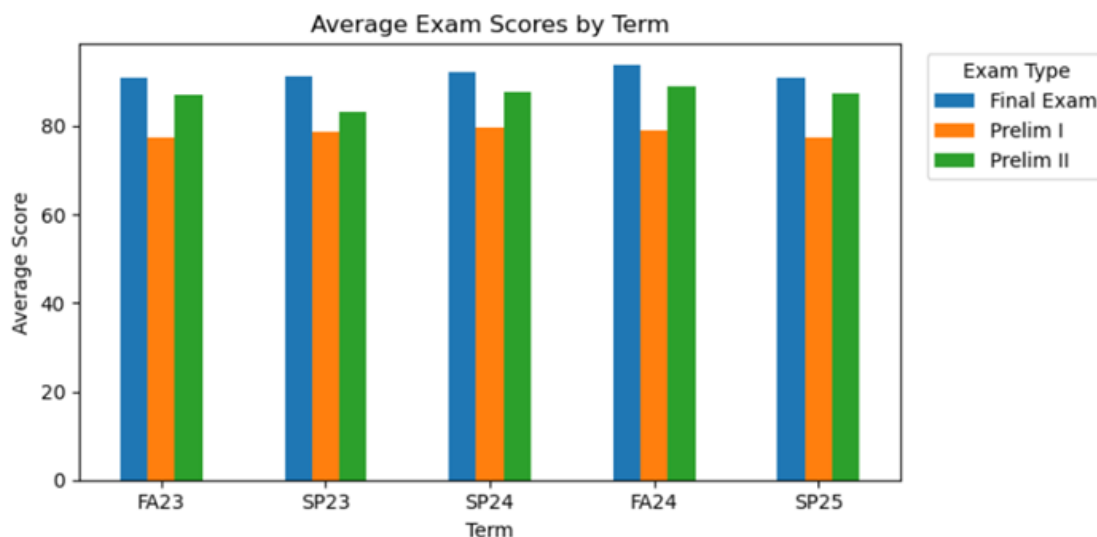


Figure 2: Average Exam Scores by Term

Exam averages remain remarkably stable across semesters.

Final Exam scores consistently outperform Prelim I across every term. Prelim II generally sits between the two. This hierarchy repeats across FA23, SP23, SP24, FA24, and SP25.

This suggests that:

- Students improve as the semester progresses.
- Instructional reinforcement between Prelim I and Final Exam is effective.
- Assessment design maintains consistent difficulty year over year.

Prelim I averages in the upper 70s to low 80s, while Finals remain in the upper 80s to low 90s. The improvement gap between Prelim I and Final is meaningful and consistent.

From an instructional standpoint, this is encouraging. It implies that early weaknesses do not permanently anchor performance. Students adjust, practice, and close gaps by the end of the term.

There is no evidence of structural grade inflation or downward drift across semesters. The exam trend line is stable, controlled, and predictable.

Average Assignment Score Over Time by Section

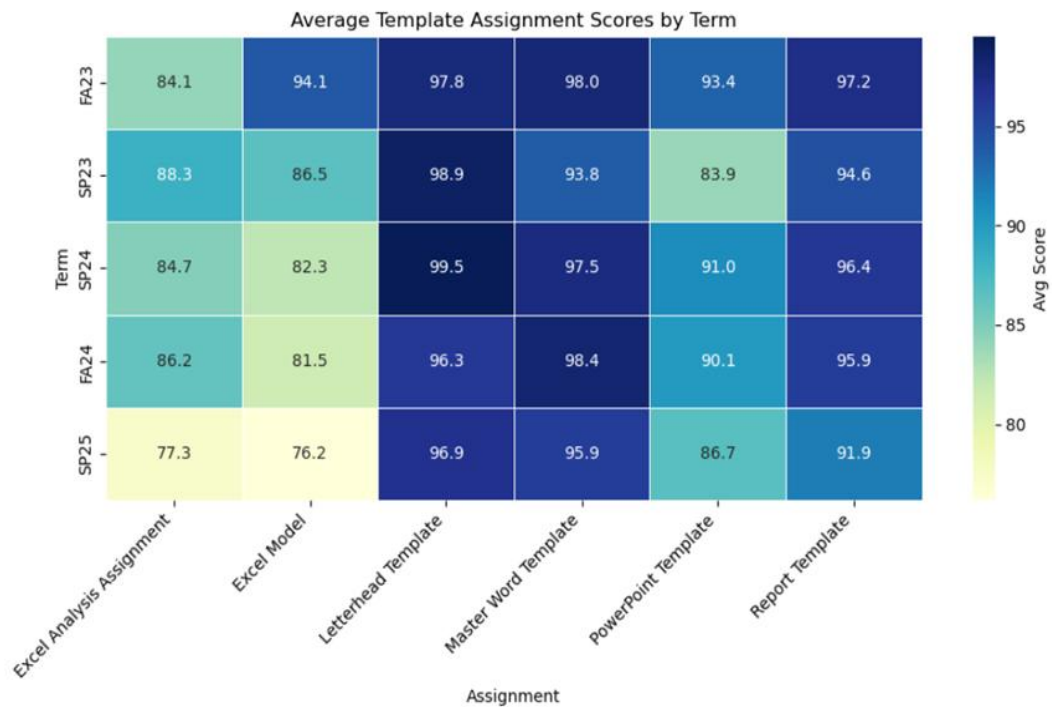


Figure 3. Average Template Assignment Scores by Term

Template based assignments demonstrate strong overall performance but reveal selective dips that are analytically important.

Letterhead and Master Word Templates consistently score above 95 across terms. These assignments are structured and procedural. Once students understand formatting conventions, performance becomes standardized.

However, Excel Model and Excel Analysis assignments show more volatility. In SP25, for example, Excel Analysis drops to the high 70s range. Excel Model scores in SP24 and FA24 trend lower than template based formatting assignments.

This confirms a recurring pattern that formatting and presentation tasks remain stable. Analytical modeling tasks introduce variability. The heatmap highlights where instructional reinforcement may be most valuable. The lower cells are not random. They cluster around analytical Excel work rather than document formatting. This distinction matters. It separates surface competency from applied analytical reasoning

Average Assignment Score Over Time by Section

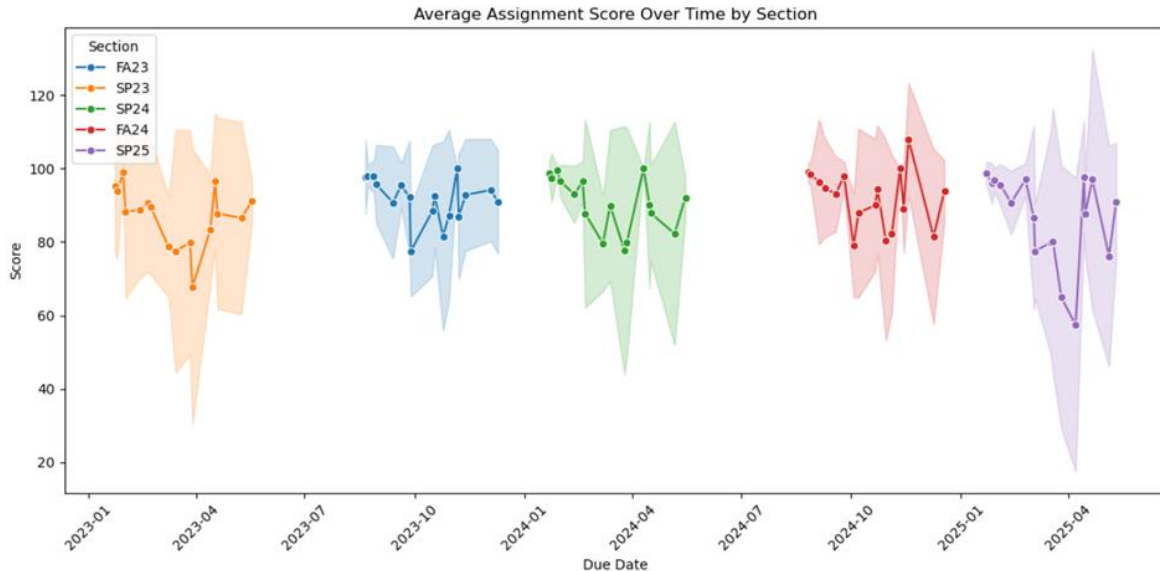


Figure 4. Average Assignment Score Over Time by Section

The time series chart reveals within term volatility and dispersion patterns. Each semester displays oscillation rather than a linear trend. Performance fluctuates around a relatively stable mean, but dispersion widens during certain weeks.

Several structural observations emerge:

- Early semester assignments often show greater variance as students adjust to expectations.
- Mid semester technical work increases dispersion, especially around Excel heavy assignments.
- End of term scores generally stabilize as expectations and grading structure become familiar.

SP25 shows wider shaded regions, indicating greater dispersion. This could reflect cohort differences, assignment design shifts, or increased difficulty in certain modules. Importantly, no term shows a systemic downward slope. The volatility is cyclical, not directional. This indicates that performance variability is assignment driven, not structural or semester driven.

Key Findings and Conclusion

Across all visualizations and term comparisons, performance patterns are structurally consistent. Students perform strongest in structured, procedural assignments such as templates and guided formatting tasks. These assignments produce averages clustered in the mid to high 90s with minimal dispersion. In contrast, analytical Excel modules and specialized function assignments consistently introduce greater variability. This dispersion is not random. It reflects the cognitive shift from following defined formatting instructions to applying logical reasoning and modeling skills.

Exam performance reinforces this structure. Prelim I averages sit in the upper 70s to low 80s range across semesters, while Final Exam averages consistently rise into the upper 80s and low 90s. This pattern holds across all analyzed terms. The improvement between early and late assessments suggests that instructional reinforcement is effective and that early underperformance does not permanently anchor outcomes. There is no visible evidence of grade inflation or downward drift. Cross term averages remain stable, indicating consistent grading rigor. Time based analysis further confirms that volatility is cyclical rather than directional. Dispersion widens during technically demanding modules and narrows as expectations stabilize later in the semester. No term exhibits systemic decline. Performance variability is assignment driven rather than cohort driven.

Taken together, the data demonstrates structural stability, predictable stress points in analytical modules, and measurable improvement over time. The primary leverage point for intervention lies in early technical reinforcement and targeted support around Prelim I. The grading framework itself does not require redesign. Instead, the opportunity lies in proactive academic support aligned with predictable variance patterns. This project transforms grade records into structural insight, allowing support strategies to be intentional rather than reactive.

Evaluation

This project's primary strength lies in its structural consistency. By standardizing multi term data into a unified warehouse, comparisons are valid and repeatable. The analysis moves beyond surface level averages and identifies where dispersion occurs, distinguishing formatting proficiency from analytical reasoning performance. Another strength is stability. The absence of structural drift across semesters reinforces confidence in grading calibration and assessment design. The framework also provides scalability. Once established, the database and ETL process can support future modeling, segmentation, and early warning systems.

However, limitations remain. The analysis relies heavily on averages, which can mask individual student trajectories and outlier behavior. While dispersion patterns are visible at the assignment level, deeper variance metrics or longitudinal student tracking could provide stronger predictive insight. The model also does not incorporate external factors such as prior coursework, academic background, or attendance behavior. Without these controls, performance interpretation remains course specific rather than fully causal.

Reflecting on the project, the motivation was rooted in personal experience. Having recently been a student, I understand how performance pressure compounds early in the semester. The data confirms that analytical Excel modules and early exams create measurable stress points. The goal was not to justify grading outcomes but to understand structure. Understanding structure allows support to be precise. Rather than broad adjustments, targeted reinforcement at predictable inflection points can materially improve outcomes.

Overall, the project demonstrates that consistent data structuring enables meaningful academic insight. The grading system itself is stable and that scores rely on the syllabus. The opportunity lies in identifying where students struggle and intervening before variance compounds.